



ReliefTrack PH:

Community Disaster and Aid Management System

Brgy. San Perfecto ReliefTrack

SDG 9 - Industry, Innovation and Infrastructure

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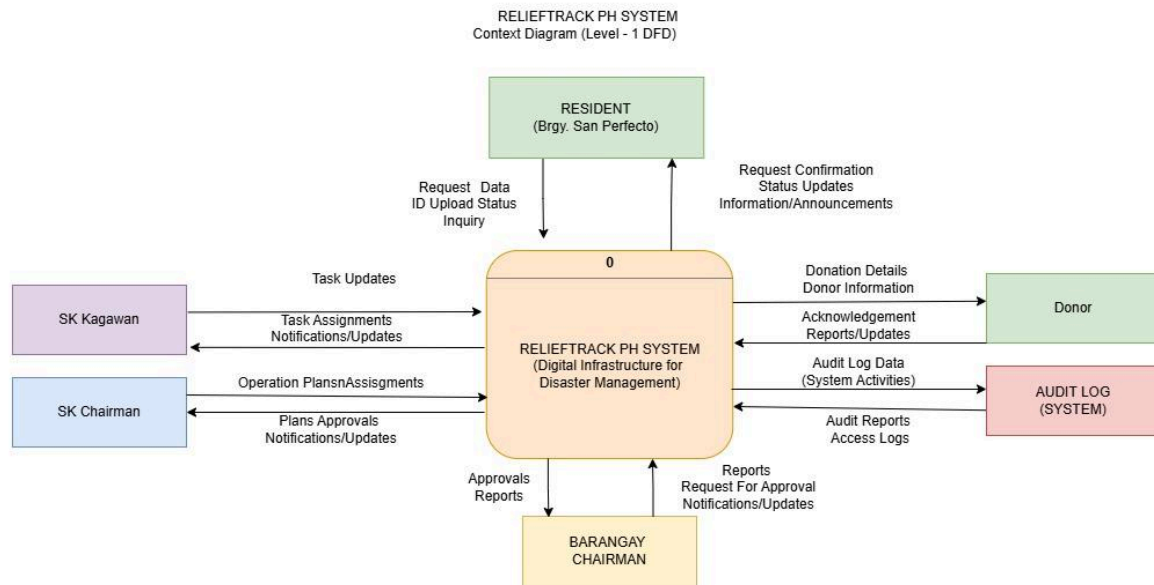
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I. Process Architecture

Level-0 Data Flow Diagram (DFD)



The Context Diagram of the **ReliefTrack PH System** shows how the system connects and communicates with different people involved in disaster management in **Brgy. San Perfecto**. The system works as the main platform where requests, reports, updates, donations, and approvals are managed during emergencies and relief operations.



1. Resident

Residents use the system to send requests for assistance, upload valid IDs, and check the status of their requests. The system then provides confirmations, updates, and announcements so residents can stay informed during disaster situations.

2. SK Kagawad

The SK Kagawad receives task assignments, notifications, and updates from the system. They also send task progress updates to help monitor ongoing activities and community support operations.

3. SK Chairman

The SK Chairman submits operation plans and assignments through the system. After reviewing, the system sends notifications, updates, and approval results to help organize disaster response activities properly.

4. Barangay Chairman

The Barangay Chairman uses the system to review reports, approval requests, and important updates. They can also send approvals and reports to support decision-making and ensure that operations are properly managed.

5. Donor

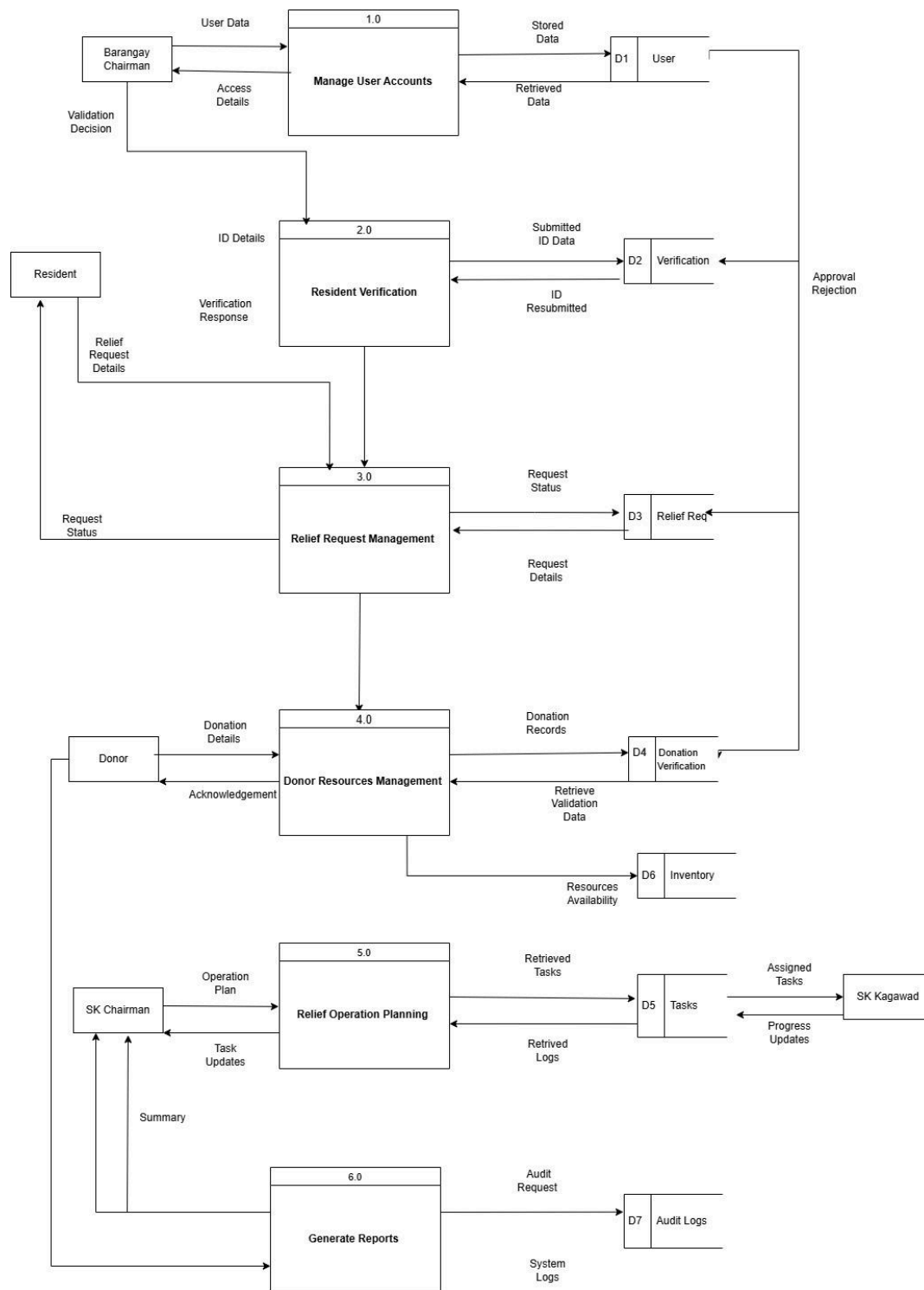
Donors can provide donation details and personal information through the system. In return, they receive acknowledgements, reports, and updates to keep them informed about their donations and how these are being used for the community.

6. Audit Log (System)

The Audit Log records important system activities and user actions. This helps maintain transparency, monitor system usage, and generate audit reports whenever needed.



1.2 Level-1 Data Flow Diagram (DFD)

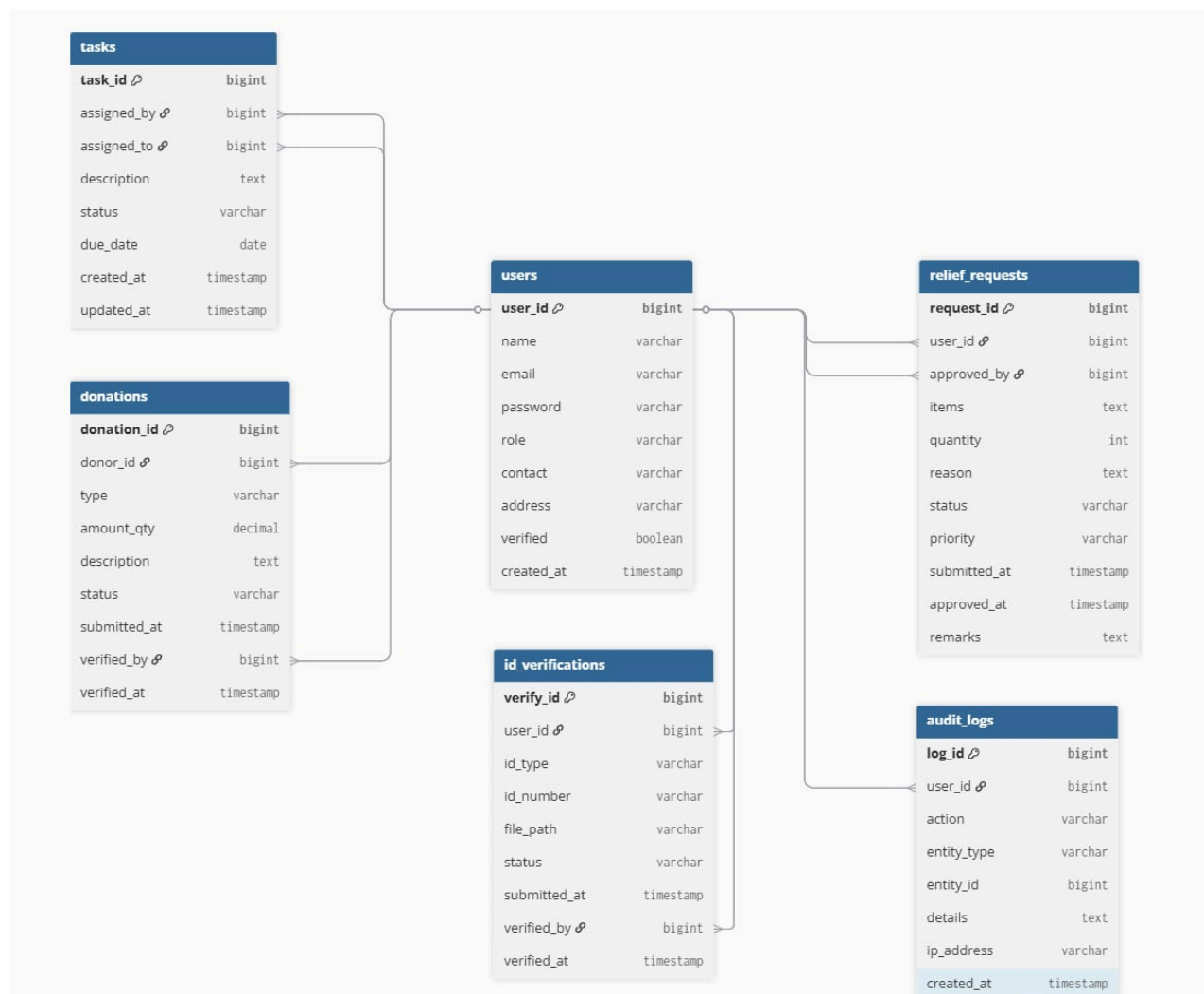




II. Data Architecture

Entity Relationship Diagram (ERD)

The ERD uses Crow's Foot notation to model six entities: users, relief_requests, tasks, donations, verifications, and audit_logs. The users table is the central entity, containing all five roles (Resident, SK Kagawad, SK Chairman, Barangay Chairman, Donor). It has one-to-many relationships with relief_requests, tasks, donations, and audit_logs, and a one-to-one relationship with verifications. The schema is normalized to 3NF with all foreign keys properly defined.





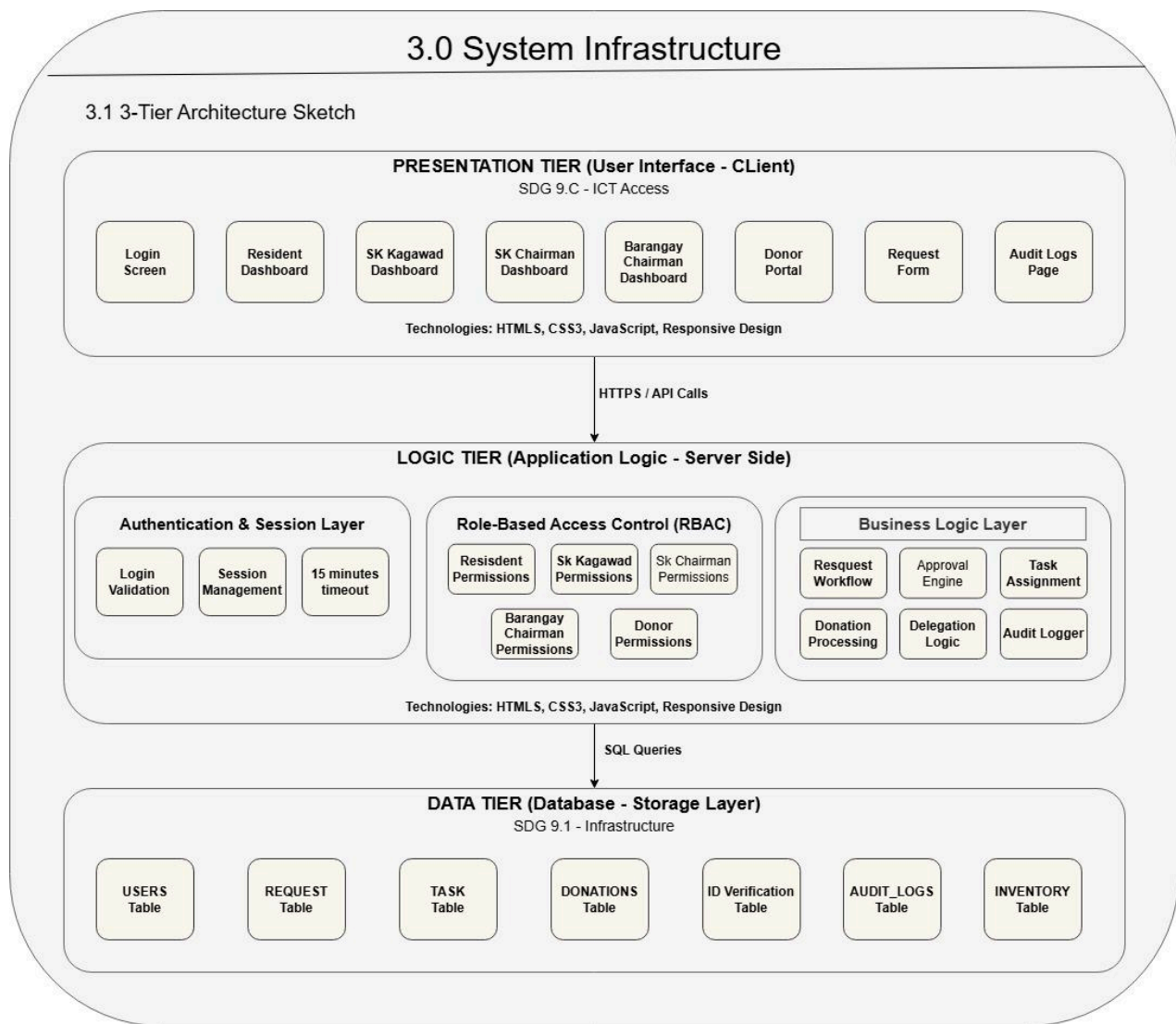
III. Relationship Cardinalities Summary

#	Relationship	Cardinality	Description
1	users → relief_requests	1 : M	One user submits many requests
2	users → tasks (assigned_to)	1 : M	One SK Kagawad receives many tasks
3	users → tasks (assigned_by)	1 : M	One SK Chairman assigns many tasks
4	users → donations (donor_id)	1 : M	One donor makes many donations
5	users → donations (verified_by)	1 : M	One chairman verifies many donations
6	users → verifications (user_id)	1 : 1	One user has one verification
7	users → verifications (verified_by)	1 : M	One chairman verifies many IDs
8	users → audit_logs	1 : M	One user generates many logs



IV. System Infrastructure

3-Tier Architecture Sketch



This diagram kinda shows the **3-Tier Architecture of our system infrastructure**, and yeah it splits into three layers: the Presentation Tier, Logic Tier, and Data Tier.

So the **Presentation Tier** is basically the user interface where people do their thing, they interact with dashboards, forms, and portals.

Then the **Logic Tier** deals with processing, authentication, permissions, and also the business logic side of it.

Lastly the Data Tier is in charge of storing, managing , and holding all system data, using database tables and that kind of arrangement.



V. TELOS Feasibility

Dimension	Assessment	Justification
Technical	HIGH	Standard web technologies (HTML, CSS, JavaScript). Self-referencing ERD can be implemented in MySQL/PostgreSQL.
Economic	HIGH	Low-cost, web-based system. Minimizes paper-based inefficiencies. No expensive licensing fees.
Legal	HIGH	Compliant with RA 10173 (Data Privacy Act). Audit trail ensures accountability.
Operational	HIGH	Role-based dashboards match existing LGU hierarchy. Minimal training required. Mobile-responsive.
Scheduling	MEDIUM	Core features: 8 weeks. Delegation feature: +2 weeks. Testing: +2 weeks. Total: 12 weeks.



VI. UI/UX Blueprint

Login Screen

ReliefTrack PH

Barangay Disaster Management • Role-Based Access

Email

you@example.com

Password

Sign in →

DEMO ACCOUNTS (TAP TO FILL)

Resident	resident@example.com / res
SK Kagawad	kagawad@example.com / kagawad
SK Chairman	skchair@example.com / skchair
Barangay Chairman	chairmen@example.com / chairman
Donor	donor@example.com / donor

Resident Dashboard

ReliefTrack PH

DISASTER MANAGEMENT

JD

Juan Dela Cruz

Resident

MENU

Dashboard

My Requests

New Request

Upload ID

Sign Out

Resident Portal

VERIFICATION

Upload ID to verify

TOTAL REQUESTS

3

All time

PENDING

1

Awaiting approval

DISTRIBUTED

1

Successfully received

Recent Requests

#5001 • Rice (5kg), Canned Goods (10)

Flood damaged home • 2024-01-16

PENDING

#5002 • Bottled Water (24)

No clean water • 2024-01-10

APPROVED

#5003 • Medicine (2 boxes)

Elderly sick • 2024-01-05

DISTRIBUTED

New Relief Request



Barangay Chairman Dashboard

ReliefTrack PH
DISASTER MANAGEMENT

MR Miguel Ramos
SK Chairman

MENU

- Dashboard
- Plan Operation
- Assign Tasks
- Monitor Volunteers
- Oversee Distribution
- Inventory Levels
- Sign Out

SK Chairman Command Center

ACTIVE OPERATION
Zone 1
Relief in progress

VOLUNTEERS
12
Deployed

TOTAL TASKS
2
1 done

★ Task Board

Distribute rice to Zone 1
Assigned to Ana Cruz • Due 2024-01-19

COMPLETED

Zone 3 Needs Assessment
Assigned to Ana Cruz • Due 2024-01-25

IN PROGRESS

Plan Operation

Assign Task

Audit Logs Page

ReliefTrack PH
DISASTER MANAGEMENT

CS Chairman Santos
Barangay Chairman

MENU

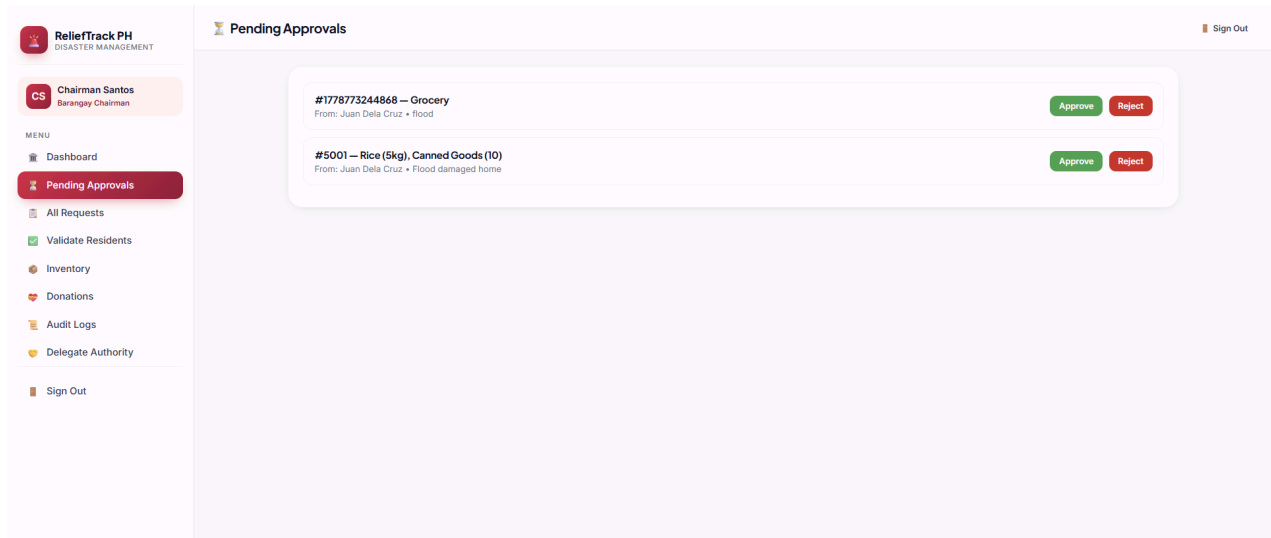
- Dashboard
- Pending Approvals
- All Requests
- Validate Residents
- Inventory
- Donations
- Audit Logs
- Delegate Authority
- Sign Out

System Audit Logs (FR-04.5 / FR-06.5)

DATE	USER	ACTION	ENTITY	DETAILS	IP
2026-05-14	Chairman Santos	LOGIN	Auth#4	User logged in	127.0.0.1
2026-05-14	Miguel Ramos	LOGOUT	Auth#3	User signed out	127.0.0.1
2026-05-14	Miguel Ramos	LOGIN	Auth#3	User logged in	127.0.0.1
2026-05-14	Juan Dela Cruz	LOGOUT	Auth#1	User signed out	127.0.0.1
2026-05-14	Juan Dela Cruz	SUBMIT_REQUEST	ReliefRequest#1778773244868	New request: Grocery	127.0.0.1
2026-05-14	Juan Dela Cruz	LOGIN	Auth#1	User logged in	127.0.0.1
2026-05-14	Ana Cruz	LOGOUT	Auth#2	User signed out	127.0.0.1
2026-05-14	Ana Cruz	UPDATE_TASK	Task#3001	Status: PENDING —	127.0.0.1
2026-05-14	Ana Cruz	LOGIN	Auth#2	User logged in	127.0.0.1
2026-05-14	Miguel Ramos	LOGOUT	Auth#3	User signed out	127.0.0.1
2026-05-14	Miguel Ramos	LOGIN	Auth#3	User logged in	127.0.0.1
2026-05-14	Juan Dela Cruz	LOGOUT	Auth#1	User signed out	127.0.0.1
2026-05-14	Juan Dela Cruz	LOGIN	Auth#1	User logged in	127.0.0.1
2024-01-11	Chairman Santos	APPROVE_REQUEST	ReliefRequest#5002	Approved with priority	192.168.1.1



Mobile Responsive View



VII. Traceability Matrix

REQUIREMENTS TRACEABILITY MATRIX (RTM)

TEST ID	REQUIREMENT TESTED	EXPECTED RESULT	STATUS
TC-01	FR-01.3 – Resident submits request	Request saved as PENDING	PASSED
TC-02	FR-01.2 – Resident uploads valid ID	ID stored as PENDING	PASSED
TC-03	FR-02.2 – SK Kagawad updates task	Task progress updated	PASSED
TC-04	FR-03.2 – SK Chairman assigns task	Task assigned successfully	PASSED
TC-05	FR-04.1 – Chairman approves request	Status changed to APPROVED	PASSED
TC-06	FR-04.5 – Chairman reviews audit logs	Audit records displayed	PASSED



TC-07	FR-05.1 – Donor submits donation	Donation saved as PENDING	PASSED
TC-08	FR-06.3 – Automatic logout after 15 minutes inactivity	Session terminated	PASSED
TC-09	FR-06.4 – Role-based access control	Unauthorized access blocked	PASSED
TC-10	FR-06.5 – Audit trail logging	Log entry generated	PASSED
TC-11	NFR-01 – Report generation speed	Reports load within 10 sec	PASSED